Package ‘DrugAdherence’

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Type Package

Title DrugAdherence: To learn patterns of Medication Compliance, Aderence and Persistence

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Description The DrugAdherence R package analyzes medication compliance and persistence in cohorts. It assesses drug utilization from first exposure, calculates key metrics like medication possession ratio and proportional days covered, and enhances insights into real-world adherence to improve patient outcomes.

Depends DatabaseConnector (>= 5.0.0),
R (>= 4.1.0)

Imports checkmate,
CirceR,
dplyr,
fabletools,
feasts,
FeatureExtraction,
ggplot2,
hrbrthemes,
viridis,
lifecycle,
rlang,
SqlRender,
tidyr,
tsibble

Suggests Eunomia,
remotes,
rmarkdown,
knitr,
RJSONIO,
testthat,
withr

VignetteBuilder knitr

Remotes CirceR

License Apache License (== 2.0)

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URL https://ohdsi.github.io/DrugAdherence/, https://github.com/OHDSI/DrugAdherence

BugReports https://github.com/OHDSI/DrugAdherence/issues

Contents

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calculateSummaryStatistics

Calculate summary statistics

Description

This function, given a data frame (df) with a continuous variable and a group, computes distribution statistics.

Usage

```
calculateSummaryStatistics(df, value = "value", group = NULL)
```

Arguments

`df`	Data frame to perform calculations on
`value`	Field in the data frame that has the data to calculate.
`group`	Any optional grouping variable

createCodeSetTableFromConceptSetExpression
Generate a Temporary Codeset Table from a Concept Set Expression

Description

This function takes a Circe-generated conceptSetExpression object (list) and creates a temporary table containing unique concept IDs.

Usage

```
createCodeSetTableFromConceptSetExpression(
  connection,
  vocabularyDatabaseSchema,
  conceptSetExpression,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  conceptSetTable = "#concept_sets"
)
```

Arguments

connection	An object of type connection as created using the connect function in the DatabaseConnector package. Can be left NULL if connectionDetails is provided, in which case a new connection will be opened at the start of the function, and closed when the function finishes.
vocabularyDatabaseSchema	Schema name where your OMOP vocabulary data resides. This is commonly the same as cdmDatabaseSchema. Note that for SQL Server, this should include both the database and schema name, for example 'vocabulary.dbo'.
conceptSetExpression	A R object (list) that is conforming to OHDSI Circe concept set expression. Usually from ROhdsiWebApi. Defines the cncept set expression as r list object. This is converted to json and given to CirceR to get concept set expression sql.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
conceptSetTable	The name of a remote temp table that has the unique concept_id. Should have column name concept_id and be unique.

createViolinPlot
Create an Enhanced Violin Plot

Description

This function generates a violin plot enhanced with overlaid boxplots and symbols for mean and median values. It uses ggplot2 for plotting, viridis for color palettes, and hrbrthemes for themes.

Usage

```
createViolinPlot(
  data,
  xName = "groupLabel",
  yName = "value",
  fillName = NULL,
  title = "Distribution Plot",
  colorMean = "black",
  colorMedian = "black",
  sizeMean = 3,
  sizeMedian = 3,
  shapeMean = 18,
  shapeMedian = 16,
  angleXLabels = 45
)
```

Arguments

`data`	A data frame containing the variables to be plotted.
`xName`	The name of the categorical variable (as a string) to plot on the x-axis.
`yName`	The name of the numeric variable (as a string) to plot on the y-axis.
`fillName`	(Optional) The name of the variable (as a string) to use for fill colors. Default is NULL.
`title`	The title of the plot. Default is "Distribution Plot".
`colorMean`	The color to use for the mean symbol. Default is "blue".
`colorMedian`	The color to use for the median symbol. Default is "green".
`sizeMean`	The size of the mean symbol. Default is 3.
`sizeMedian`	The size of the median symbol. Default is 3.
`shapeMean`	The shape of the mean symbol. Default is 18.
`shapeMedian`	The shape of the median symbol. Default is 16.
`angleXLabels`	What is angle for the x tick labels.

Value

A ggplot object representing the violin plot.

`getDrugAdherenceInDenominatorCohort`

get drug exposure events for a person

Description

Given a concept set expression and a denominator cohort to restrict the persons and exposure event to period in the denominator cohort, this function creates a temp table that has the records from drug_exposure table.

Usage

```
getDrugAdherenceInDenominatorCohort(
  connection = NULL,
  querySource = TRUE,
  cdmDatabaseSchema,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  conceptSetTable = "#concept_sets",
  denominatorCohortDatabaseSchema = NULL,
  denominatorCohortTable = "#denominator",
  denominatorCohortId = 0,
  forceMinimumDaysSupply = 1,
  DrugAdherenceOutputTable = "#drug_exposure",
  restrictToCohortPeriod = TRUE
)
```

Arguments

`connection` An object of type `connection` as created using the `connect` function in the DatabaseConnector package. Can be left NULL if `connectionDetails` is provided, in which case a new connection will be opened at the start of the function, and closed when the function finishes.

`querySource` In the drug_exposure table do you want to query drug_source_concept_id. This is set to TRUE by default.

`cdmDatabaseSchema` Schema name where your patient-level data in OMOP CDM format resides. Note that for SQL Server, this should include both the database and schema name, for example 'cdm_data.dbo'.

`tempEmulationSchema` Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

`conceptSetTable` The name of a remote temp table that has the unique concept_id. Should have column name concept_id and be unique.

`denominatorCohortDatabaseSchema` The cohort database schema that has the denominator cohort.

`denominatorCohortTable` The cohort table name that has the denominator cohort.

`denominatorCohortId` The cohort id of the denominator cohort.

`forceMinimumDaysSupply` (Default 1, i.e. not used). Acceptable values are NULL, or any integer value to represent days.

`DrugAdherenceOutputTable` the output table

`restrictToCohortPeriod` Do you want to restrict to cohort period? Default = TRUE.

getNumeratorCohorts *Create Numerator Cohorts*

Description

This function takes an input a cohort (called denominator to bind the numerator cohort to), a concept set expression to create the numerator cohort.

Usage

```
getNumeratorCohorts(
  connection = NULL,
  cdmDatabaseSchema,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  numeratorCohortTableBaseName = "#numerator",
  DrugAdherenceTable = "#drug_exposure",
  gapDays = c(0),
  baseCohortDefinitionId = 100
)
```

Arguments

`connection`	An object of type `connection` as created using the `connect` function in the DatabaseConnector package. Can be left NULL if `connectionDetails` is provided, in which case a new connection will be opened at the start of the function, and closed when the function finishes.
`cdmDatabaseSchema`	Schema name where your patient-level data in OMOP CDM format resides. Note that for SQL Server, this should include both the database and schema name, for example 'cdm_data.dbo'.
`tempEmulationSchema`	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
`numeratorCohortTableBaseName`	The name of the output table
`DrugAdherenceTable`	The name of the table with "#drug_exposure". Should be a temp table.
`gapDays`	(optional) Number of days to check for persistence of drug exposure. Can take a array of days, and report will be generated for all. This is also called persistence in Atlas, or collapse era days in cohort definition.
`baseCohortDefinitionId`	The minimum cohortId to create cohorts for all gapDays

processTimeSeries *Process Dates for STL Time Series Analysis*

Description

This function performs STL decomposition on a data frame for specified time representations. It requires a date field and can optionally use a weight field to aggregate data. The function handles different time aggregations (e.g., Day, Month, Quarter, Year) and returns a list of models, components, and plots for each time representation.

Usage

```
processTimeSeries(
  df,
  timeRepresentations = c("Day", "Week", "Month", "Quarter", "Year"),
  dateField = "cohortStartDate",
  weight = NULL
)
```

Arguments

`df`	A data frame containing the time series data.
`timeRepresentations`	A character vector specifying the time units to aggregate by. Accepted values are 'Day', 'Month', 'Quarter', 'Year'. Default is c('Day', 'Month', 'Quarter', 'Year').
`dateField`	The name of the column in `df` that contains the date information. Defaults to 'cohortStartDate'.
`weight`	An optional name of the column that contains weights for aggregation. If NULL, each record is assumed to have an equal weight of 1.

Value

A list where each element corresponds to a specified time representation and contains the processed data, the fitted model, its components, and a plot of the decomposition.

runDrugAdherence *Run Drug Exposure Analysis*

Description

This function takes as a input a Circe compatible concept set expression (as r list object that can be converted to json), a denominator cohort, and checks for occurrence of drug exposure events in the drug_exposure table of the CDM for the conceptId's in the resolved set of the given concept expression in the period and for the subjects in the denominator cohort. It then computes a series of drug utilization metrics (adherence, persistence, utilization patterns) and reports returns a list of objects that maybe utilized in a drug exposure report.

Usage

```
runDrugAdherence(
  connectionDetails = NULL,
  connection = NULL,
  conceptSetExpression,
  querySource = TRUE,
  cdmDatabaseSchema,
  denominatorCohortDatabaseSchema,
  denominatorCohortTable,
  denominatorCohortId,
  forceMinimumDaysSupply = 1,
  maxFollowUpDays = 365,
  vocabularyDatabaseSchema = cdmDatabaseSchema,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  gapDays = c(0)
)
```

Arguments

`connectionDetails`
: An object of type `connectionDetails` as created using the `createConnectionDetails` function in the DatabaseConnector package. Can be left NULL if `connection` is provided.

`connection`
: An object of type `connection` as created using the `connect` function in the DatabaseConnector package. Can be left NULL if `connectionDetails` is provided, in which case a new connection will be opened at the start of the function, and closed when the function finishes.

`conceptSetExpression`
: A R object (list) that is conforming to OHDSI Circe concept set expression. Usually from ROhdsiWebApi. Defines the cncept set expression as r list object. This is converted to json and given to CirceR to get concept set expression sql.

`querySource`
: In the drug_exposure table do you want to query drug_source_concept_id. This is set to TRUE by default.

`cdmDatabaseSchema`
: Schema name where your patient-level data in OMOP CDM format resides. Note that for SQL Server, this should include both the database and schema name, for example 'cdm_data.dbo'.

`denominatorCohortDatabaseSchema`
: The cohort database schema that has the denominator cohort.

`denominatorCohortTable`
: The cohort table name that has the denominator cohort.

`denominatorCohortId`
: The cohort id of the denominator cohort.

`forceMinimumDaysSupply`
: (Default 1, i.e. not used). Acceptable values are NULL, or any integer value to represent days.

`maxFollowUpDays`
: (optional) Number of days to follow the persons in the denominator cohort.

`vocabularyDatabaseSchema`
: Schema name where your OMOP vocabulary data resides. This is commonly the same as cdmDatabaseSchema. Note that for SQL Server, this should include both the database and schema name, for example 'vocabulary.dbo'.

tempEmulationSchema
: Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

gapDays
: (optional) Number of days to check for persistence of drug exposure. Can take a array of days, and report will be generated for all. This is also called persistence in Atlas, or collapse era days in cohort definition.

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runFeatureExtraction *Run feature extraction*

Description

Run feature extraction

Usage

```
runFeatureExtraction(
  connectionDetails = NULL,
  connection = NULL,
  cdmDatabaseSchema,
  cohortDatabaseSchema,
  cohortIds,
  cohortTable,
  covariateSettings,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  outputFolder = NULL,
  aggregated = TRUE
)
```

Arguments

connectionDetails
: An object of type `connectionDetails` as created using the `createConnectionDetails` function in the DatabaseConnector package. Can be left NULL if `connection` is provided.

connection
: An object of type `connection` as created using the `connect` function in the DatabaseConnector package. Can be left NULL if `connectionDetails` is provided, in which case a new connection will be opened at the start of the function, and closed when the function finishes.

cdmDatabaseSchema
: Schema name where your patient-level data in OMOP CDM format resides. Note that for SQL Server, this should include both the database and schema name, for example 'cdm_data.dbo'.

cohortDatabaseSchema
: Schema name where your cohort table exists. cohort table should have columns cohort_definition_id, subject_id, cohort_start_date, cohort_end_date. The dates for the same subject_id, cohort_definition_id should be non overlapping. Note that for SQL Server, this should include both the database and schema name, for example 'cdm_data.dbo'.

`cohortIds` cohort ids to run feature extraction on. Only one cohort id if aggregated = FALSE.

`cohortTable` cohort Table Name.

`covariateSettings`
See FeatureExtraction covariateSettings.

`tempEmulationSchema`
Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

`outputFolder` Name of local folder to place results; make sure to use forward slashes (/). Do not use a folder on a network drive since this greatly impacts performance.

`aggregated` See `aggregated` in FeatureExtraction:getDbCovariateData.

Details

This function executes feature extraction

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